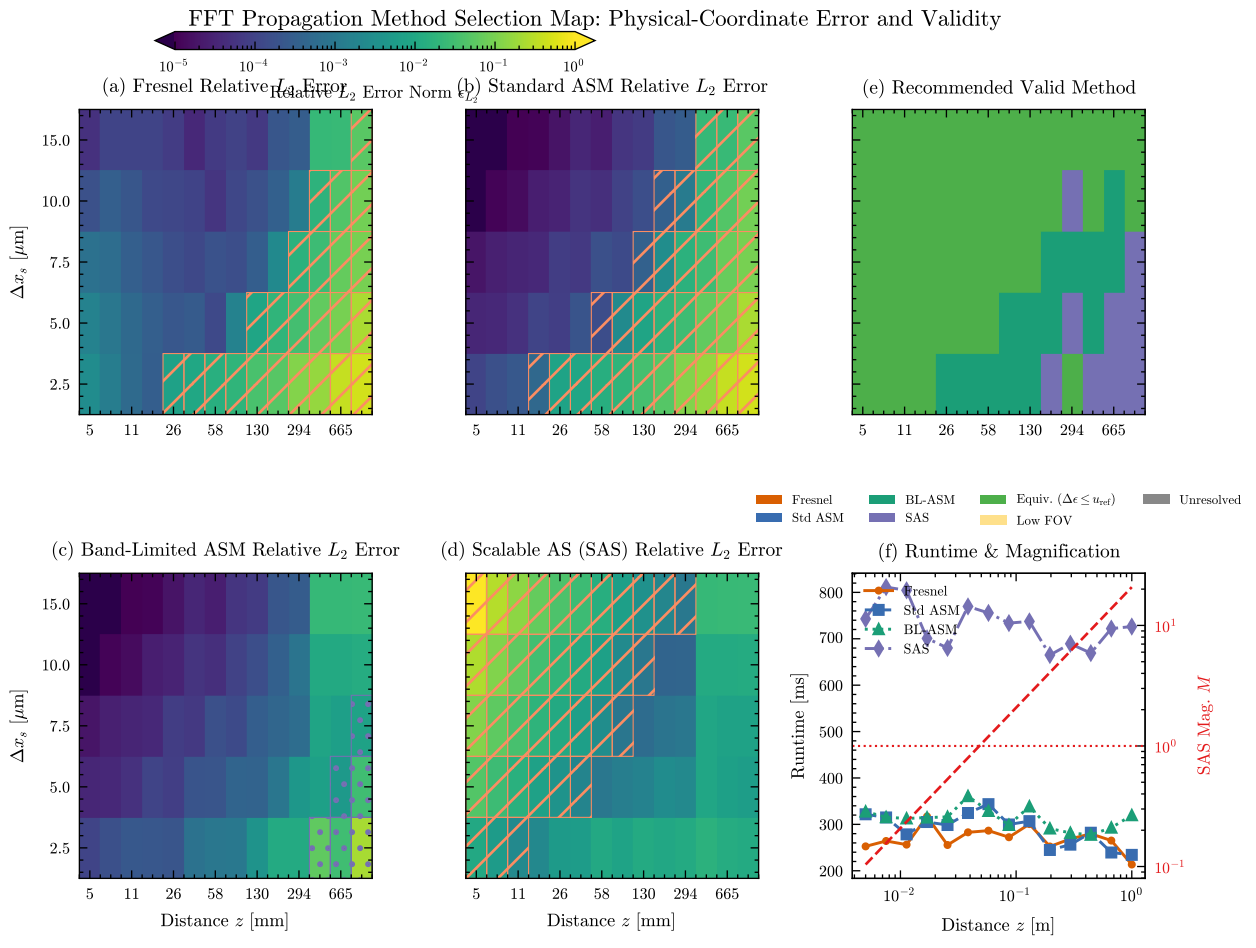

FFT Propagation Method Selection Map: Physical-Coordinate Error and Validity
Relative L_2 Error Norm ϵ_{L_2}
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0}
(a) Fresnel Relative L_2 Error
(b) Standard ASM Relative L_2 Error
(e) Recommended Valid Method
(c) Band-Limited ASM Relative L_2 Error
(d) Scalable AS (SAS) Relative L_2 Error
(f) Runtime & Magnification
Δx_s [μm]
15.0
10.0
7.5
5.0
2.5
5 11 26 58 130 294 665
Distance z [mm]
Fresnel
Std ASM
BL-ASM
SAS
Equiv. ($\Delta\epsilon \leq u_{\text{ref}}$)
Low FOV
Unresolved
Runtime [ms]
800 700 600 500 400 300 200
SAS Mag. M
10^{1} 10^{0} 10^{-1}
10^{-2} 10^{-1} 10^{0}
Distance z [m]